

Warden: solution write-up

Warden is cloud cleanup that proves it's safe first. It is an AI agent built on TrueForge for *Agents That Act* (TrueFoundry × Polaris, 26 September 2026).

The problem

About 29% of cloud spend is wasted (Flexera 2026). Finding idle resources is easy. Deleting them safely is not: a cleanup script given the wrong IDs deleted 883 Atlassian customer sites. Teams stall on fear, recurring waste and change paperwork.

What the agent reaches

Through our Python MCP server (boto3), the agent reaches:

- AWS **EC2**: volumes, snapshots, instances and Elastic IPs
- **CloudWatch**, **CloudTrail**, the **Recycle Bin**, **Route 53** and **ELBv2**
- A **Daytona sandbox**, where it runs Warden's proof script to recompute the savings independently

Where it stops

- It refuses production, legal-hold and IaC/autoscaling resources, anything an image, DNS record or load balancer still uses, and prompt-injection text in tags.
- Every change needs an independent Watchdog's single-use, HMAC-signed sign-off **and** a human Allow in TrueForge.
- Permanent steps (IP release) come only after a quarantine window, and need their own approval.
- It never terminates instances, never touches KMS keys and never deletes S3 buckets.

Architecture

The flow is TrueForge chat → Warden MCP server → AWS. Inside the server:

- **Scanner**: builds a locked plan.
- **Watchdog**: re-checks and signs.
- **Executor**: the only code that mutates AWS.
- **Ledger**: hash-chained.

A read-only Warden Console shows live state and undo countdowns. Actions are reversible first: backups, the Recycle Bin, stop instead of terminate.

How TrueForge was used

- A custom MCP connector (19 tools) with approval on all 10 action tools
- The Daytona sandbox
- Generative-UI dashboards
- The agent defined in code with the SDK
- A model through the TrueFoundry AI Gateway
- Sessions

Real versus mocked

TrueForge, the AI Gateway model, the Daytona sandbox and all of Warden's code are real. Our AWS account never activated (`OptInRequired`), so the demo uses **moto**, a local AWS simulator, and the screen says so. Switching to real AWS is one line in `.env`.

Known limits

- One region and one account per run
- Four resource types
- Idle means "quiet in the look-back window"
- List-price savings estimates
- A tamper-evident, not tamper-proof, ledger

The repo has 491 automated tests and an MIT licence.